11.3.1 Controlling access to the configuration

Access to the ElastiCache service is controlled with the help of the IAM service. The IAM service is responsible for controlling access to actions like creating, updating, and deleting a cache cluster. IAM doesn't manage access inside the cache; that's the job of the cache engine. An IAM policy defines the configuration and management actions a user, group, or role is allowed to execute on the ElastiCache service. Attaching the IAM policy to IAM users, groups, or roles controls which entity can use the policy to configure an ElastiCache cluster. You can get a complete list of IAM actions and resource-level permissions supported at http://mng.bz/WM9x.

SECURITY WARNING It's important to understand that you don't control access to the cache nodes using IAM. Once the nodes are created, security groups control the access on the network layer. Redis optionally supports user authentication and authorization.

11.3.2 Controlling network access

Network access is controlled with security groups. Remember the security group from the minimal CloudFormation template in section 11.1 where access to port 6379 (Redis) was allowed for all IP addresses. But because cluster nodes have only private IP addresses this restricts access to the VPC, as shown here:

```
Resources:
  # [...]
  CacheSecurityGroup:
    Type: 'AWS::EC2::SecurityGroup'
    Properties:
      GroupDescription: cache
      VpcId: !Ref VPC
      SecurityGroupIngress:
      - IpProtocol: tcp
        FromPort: 6379
        ToPort: 6379
        CidrIp: '0.0.0.0/0'
```

You should improve this setup by working with two security groups. To control traffic as tightly as possible, you will not allow traffic from certain IP addresses. Instead, you create two security groups. The client security group will be attached to all EC2 instances communicating with the cache cluster (your web or application servers). The cache cluster security group allows inbound traffic on port 6379 only for traffic that comes from the client security group. This way you can have a dynamic fleet of